

Introduction to LangChain

1 Introduction

This video introduces **LangChain** and explains why it is important for building applications using Large Language Models (LLMs). The focus is on moving from simple prompts to structured AI applications.

Example: Instead of just asking ChatGPT a question manually, you build a complete app that answers questions from your PDF documents automatically.

2 What is LangChain?

LangChain is a framework that helps developers build applications powered by LLMs by connecting models with data sources, tools, memory, and workflows. It acts as an orchestration layer around LLMs.

Example: You connect an LLM with a database and allow users to ask natural language questions like: “Show me total sales from last month,” and the system converts it into SQL automatically.

3 Why Do We Need LangChain?

Raw LLM APIs (like OpenAI models) only generate text. Real-world apps require memory, document retrieval, tool usage, and multi-step reasoning — which LangChain manages efficiently.

Without LangChain, you would manually handle:

- Prompt formatting
- Context injection
- Memory storage
- API chaining

Example: Building a chatbot that remembers previous messages requires state management — LangChain provides built-in memory components.

4 High-Level Discussion of an AI App Architecture

A typical LangChain app includes:

- **LLM** → Generates responses

- **Prompt Templates** → Structured input formatting
- **Memory** → Stores conversation history
- **Chains** → Sequence of operations
- **Agents** → Decide which tools to use
- **External Tools** → APIs, databases, search engines

This architecture allows modular, scalable AI systems.

Example:

User asks: “Summarize today’s tech news.”

Agent → Uses search tool → Retrieves articles → Sends to LLM → Returns summarized output.

5 Benefits of LangChain

LangChain provides:

- Modular architecture
- Reusability of components
- Easy integration with external tools
- Support for RAG (Retrieval Augmented Generation)
- Simplified production-level AI pipelines

It reduces boilerplate code and makes LLM apps production-ready.

Example: Instead of writing complex logic for document retrieval and context injection, you use LangChain’s built-in retriever and chain classes.

6 What Can You Build with LangChain?

With LangChain, you can build:

- AI chatbots with memory
- PDF question-answering systems
- AI agents that use tools
- Code assistants
- SQL query generators
- Knowledge-base assistants

Example: Upload a company policy PDF → Ask: “What is the leave policy?” → System retrieves relevant paragraph → LLM answers precisely.

7 Alternatives to LangChain

Other frameworks for LLM application development include:

- LlamaIndex – Focused on data indexing and retrieval
- Haystack – Popular for search and QA systems
- Direct API usage (e.g., raw OpenAI API)

LangChain is strong in orchestration and agent-based workflows, while alternatives may specialize more in retrieval or search pipelines.

Example:

If your main goal is building a document Q&A system, LlamaIndex may be simpler. If you need multi-step reasoning with tool usage, LangChain is more powerful.
